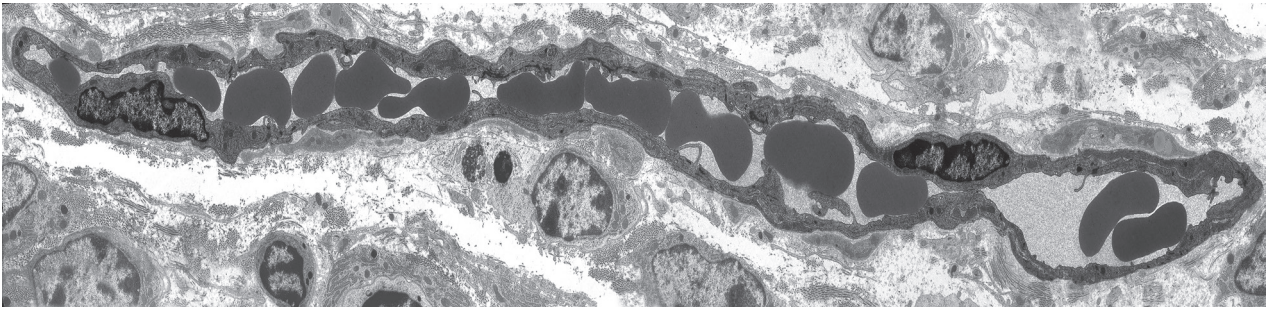


5 Fig. 5.1 is a longitudinal section of a capillary in muscle tissue as viewed with a transmission electron microscope.



magnification  $\times 2000$

**Fig. 5.1**

(a) State the evidence, **visible in Fig. 5.1**, that identifies the cells inside the capillary as red blood cells.

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..... [2]

(b) Explain how the structure of the capillary wall is related to its functions.

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- The diagram illustrates the transport of carbon dioxide (CO<sub>2</sub>) in a red blood cell. On the left, CO<sub>2</sub> enters the cell from the tissues, while O<sub>2</sub> exits. Inside the cell, CO<sub>2</sub> and H<sub>2</sub>O react, catalyzed by the enzyme carbonic anhydrase, to form carbonic acid (H<sub>2</sub>CO<sub>3</sub>). This acid then dissociates into bicarbonate ions (HCO<sub>3</sub><sup>-</sup>) and hydrogen ions (H<sup>+</sup>). HCO<sub>3</sub><sup>-</sup> is transported out of the cell, and O<sub>2</sub> enters the cell. On the right, CO<sub>2</sub> exits the cell into the tissues, and O<sub>2</sub> enters the cell from the tissues.

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- Explain how an increase in carbon dioxide in the blood leads to the release of more oxygen from red blood cells.

[4]

- (ii) Chloride ions are a constituent of blood plasma. The concentration of chloride ions in the plasma of deoxygenated blood is between  $2\text{--}4\text{ mmol dm}^{-3}$  lower than in the plasma of oxygenated blood.

Explain why the concentration of chloride ions in the blood plasma of deoxygenated blood is lower than in the plasma of oxygenated blood.

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..... [2]

[Total: 11]